Ordered Pairs in All Four Quadrants

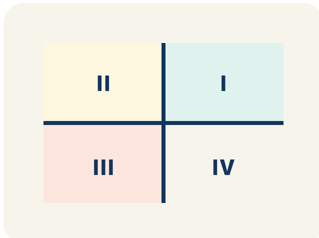
Lesson 9-5

Name: ______________________ Date: ______________________ Class: ______________________

Key Vocabulary

Level 1 support

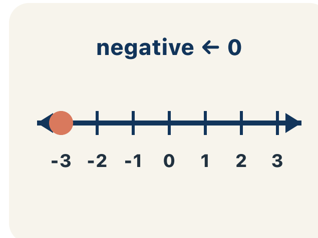
Picture first, then the word, then a plain-language meaning. Say each word out loud.



I (+, +) top-right, II (-, +) top-left, III (-, -) bottom-left, IV (+, -) bottom-right

Quadrant

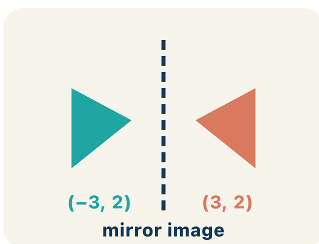
One of the four parts of a coordinate grid, numbered I to IV.



(-3, 2) means 3 left and 2 up from the origin

Negative coordinate

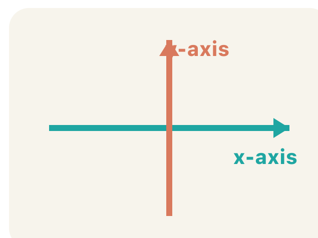
A coordinate less than zero. It means left or below the center point.



(3, 2) reflected over the y-axis is (-3, 2) — same distance, opposite side

Reflection

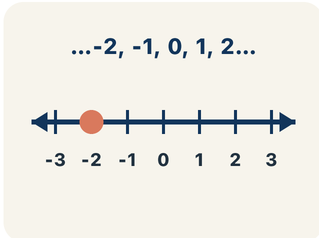
A flipped, mirror image of a point across a line.



The x-axis goes left-right; the y-axis goes up-down; they cross at (0, 0)

Axis

A line on the grid. The x-axis goes across; the y-axis goes up.



..., -3, -2, -1, 0, 1, 2, 3, ...

Integer

Whole numbers and their opposites, like -2, -1, 0, 1, 2.

Key Ideas & Notes

🎯 **WHAT WE'RE LEARNING TODAY**

I can plot ordered pairs in all four quadrants of the coordinate plane.

1 Notice

Captain Vega realized her treasure map only covered one corner of the island!

2 Key idea

I can plot ordered pairs in all four quadrants of the coordinate plane.

3 Words I'll use

Quadrant

Negative coordinate

Reflection **Axis**

Integer

🤔 Think About It

- How do you know which quadrant a point is in by looking at the signs of its coordinates?
- What do negative x-values and negative y-values mean on the map?
- Where is the point (0, 0) on the expanded map?

See the notes in action: watch one worked all the way through, then try the next with the same steps.

👀 I do — watch

Follow each step as your teacher solves it.

Problem: In which quadrant is the point (-4, 5)?

A. Quadrant I

B. Quadrant II

C. Quadrant III

D. Quadrant IV

Step 1 The x-coordinate is negative (left) and the y-coordinate is positive (up).

Step 2 Left and up is Quadrant II.

✅ **Answer:** B. Quadrant II

🤝 We do — together

Solve this one with your class using the same steps.

Problem: A point has a positive x-coordinate and a negative y-coordinate. Which quadrant is it in?

A. Quadrant I

B. Quadrant II

C. Quadrant III

D. Quadrant IV

Step 1 ______________________________

Step 2 ______________________________

Answer: ______________________________

✏️ You do — your turn

Now try one on your own. Show every step.

Problem: Which point is located on the x-axis?

A. (4, 0)

B. (0, 4)

C. (4, 4)

D. (-4, -4)

Show your work:

____

____

____

Turn & Talk — Discussion Points

Level 1 support | Level 2

Talk with a partner about each prompt. If you need help getting started, use the Level 1 sentence stems. Ready for more? Try the Level 2 question.

LAUNCH

What do the signs of the coordinates in an ordered pair tell you about which quadrant the point is in?

Level 1 support

Start here: The signs of the coordinates tell me the quadrant because ____.

💡 **Need a hint?**

Sentence starters (optional)

If both coordinates are positive, the point is in quadrant ____.

Si ambas coordenadas son positivas, el punto esta en el cuadrante ____.

If the x is negative and the y is positive, the point is in quadrant ____.

Si la x es negativa y la y es positiva, el punto esta en el cuadrante ____.

Word bank: quadrant | ordered pair | negative coordinate | x-axis | y-axis

Listen for: *Student matches sign combinations to quadrants (e.g., (+,+) is I, (-,+) is II).*

Level 2

Push students to explain how to tell which quadrant a point is in without plotting it.

- I can find the quadrant without plotting by ____.
- The signs of (x, y) tell me ____.

EXPLORE

Plot (-3, -4). Which quadrant is it in, and how do the negative coordinates change the directions you move from the origin?

Level 1 support

Start here: Negative coordinates change my moves because ____.

💡 **Need a hint?**

Sentence starters (optional)

A negative x means I move ____.

Una x negativa significa que me muevo ____.

A negative y means I move ____.

Una y negativa significa que me muevo ____.

Word bank: **quadrant** **negative coordinate** **ordered pair** **x-axis** **y-axis**

Listen for: *Student moves left for negative x and down for negative y, landing in quadrant III.*

Level 2

Ask students to predict the quadrant of (-3, 4) and compare how it differs from (-3, -4).

- (-3, 4) is in quadrant ____, but (-3, -4) is in quadrant ____ because ____.
- Changing the sign of y moves the point from ____ to ____.

CONNECT

How are the four quadrants like a tic-tac-toe of positive and negative directions on the coordinate plane?

Level 1 support

Start here: Each quadrant is a combination of ____ and ____.

💡 **Need a hint?**

Sentence starters (optional)

Quadrant ____ has both coordinates positive.

El cuadrante ____ tiene ambas coordenadas positivas.

Quadrant ____ has both coordinates negative.

El cuadrante ____ tiene ambas coordenadas negativas.

Word bank: quadrant | negative coordinate | ordered pair | x-axis | y-axis

Listen for: *Student connects each quadrant to a unique sign pattern for x and y.*

Level 2

Push students to generalize what happens to a point that lands exactly on an axis instead of inside a quadrant.

- A point on the x-axis has a y-coordinate of ____.
- A point on an axis is not in any quadrant because ____.

REFLECT

A point has coordinates (5, -2). Without plotting, how can you describe exactly where it is and which quadrant it is in?

Level 1 support

Start here: The point (5, -2) is in quadrant ____ because ____.

💡 **Need a hint?**

Sentence starters (optional)

The point is ____ of the y-axis and ____ of the x-axis.

El punto esta a la ____ del eje y y ____ del eje x.

It is in quadrant ____ because the signs are ____.

Esta en el cuadrante ____ porque los signos son ____.

Word bank: quadrant | ordered pair | negative coordinate | x-axis | y-axis

Listen for: *Student says (5, -2) is right and down, in quadrant IV, from the signs alone.*

Level 2

Ask students to critique a classmate who said (5, -2) is in quadrant I and explain the error.

- The classmate is wrong because ____.
- The negative y-coordinate means the point must be ____.

Write About the Math

The Writing Revolution

I can explain my work using the words quadrant, negative coordinate, reflection, and axis.

1. Kernel Sentence

subject + verb

Model: Quadrant is one of the four parts of a coordinate grid, numbered I to IV.
Cuadrante es una de las cuatro partes de una cuadrícula de coordenadas, numeradas I a IV.

Write a kernel sentence about quadrant. Use a subject and a verb.
Escribe una oración base sobre cuadrante. Usa un sujeto y un verbo.

____

____

2. Sentence Expansion

because · but · so

Kernel: Quadrant matters in math
Cuadrante importa en matemáticas

Expand the kernel three ways. Add a reason, a contrast, and a result.

because / ***porque***

Quadrant matters in math because ____.
Cuadrante importa en matemáticas porque ____.

____

but / ***pero***

Quadrant matters in math, but ____.
Cuadrante importa en matemáticas, pero ____.

____

so / ***entonces***

Quadrant matters in math, so ____.
Cuadrante importa en matemáticas, entonces ____.

____

3. Sentence Types

4 ways to write a math idea

Statement
Afirmación

Tell one true fact about quadrant.

Di un hecho verdadero sobre quadrant.

Quadrant ____.

Question
Pregunta

Ask a question about quadrant.

Haz una pregunta sobre quadrant.

How does ____ ?

¿Cómo ____ ?

Exclamation
Exclamación

Show excitement about quadrant.

Muestra entusiasmo sobre quadrant.

Wow, ____ !

¡Guau, ____ !

Command
Mandato

Tell a partner what to do with quadrant.

Dile a un compañero qué hacer con quadrant.

First, ____ .

Primero, ____ .

4. Explain Your Reasoning

use a sentence starter

If both coordinates are positive, the point is in quadrant ____.

Si ambas coordenadas son positivas, el punto esta en el cuadrante ____.

If the x is negative and the y is positive, the point is in quadrant ____.

Si la x es negativa y la y es positiva, el punto esta en el cuadrante ____.

A negative x means I move ____.

Una x negativa significa que me muevo ____.

Try It

Solve on your own. Check the answer key when you are done.

1. Which ordered pair is in Quadrant IV?

A. (8, -3)
B. (-8, 3)
C. (8, 3)
D. (-8, -3)

Show your work:

2. The point (0, -7) lies on which axis?

A. The y-axis
B. The x-axis
C. Quadrant III
D. Quadrant IV

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

A point starts in Quadrant I at (4, 6). Describe a move that puts it in Quadrant III. What must happen to both coordinates? Can you give the exact new coordinates after your move?

Sentence starter: To move from Quadrant I to Quadrant III, both coordinates must become ____. Starting at (4, 6), I could move ____ units left and ____ units down to reach (____,____), which is in Quadrant III because ____.

Show your work:

Reflect — Exit Ticket

A point has a negative x-coordinate and a negative y-coordinate. Which quadrant is it in?

A. Quadrant I
B. Quadrant II
C. Quadrant III
D. Quadrant IV

Your answer:

Answer Key & Teacher Guide

1. **We Try (notes):** D. Quadrant IV *— Positive x (right) and negative y (down) = Quadrant IV (bottom-right).*
2. **You Try (notes):** A. (4, 0) *— A point is on the x-axis when its y-coordinate is 0. (4, 0) is on the x-axis.*
3. **Try It 1:** A. (8, -3) *— Quadrant IV has positive x and negative y: (8, -3).*
4. **Try It 2:** A. The y-axis *— With x = 0, the point is on the y-axis.*
5. **Exit Ticket:** C. Quadrant III *— Quadrant III is where both x and y are negative. This means the point is left of the origin and below the origin.*

Writing (TWR) — what to look for

- **Kernel sentence:** A complete sentence needs a subject and a verb. Example: Quadrant is one of the four parts of a coordinate grid, numbered I to IV.
- **Expansion:** *because* gives a reason, *but* shows a contrast or exception, *so* shows a result. Answers vary; each must keep the kernel idea and add the correct kind of detail.
- **Sentence types:** Statement ends with a period, question with "?", exclamation with "!", and a command starts with an action verb (a "bossy" verb).